

■ 3.7.2 Getting Pieces of Matrices

$m[[i, j]]$	the i, j^{th} entry
$m[[i]]$	the i^{th} row
$\text{Transpose}[m][[i]]$ or $\text{Map}[\#[[i]]\&, m]$	the i^{th} column
$m[[\{i_1, \dots, i_r\}, \{j_1, \dots, j_s\}]]$	the $r \times s$ submatrix of m with elements having row indices i_k and column indices j_k
$m[[\text{Range}[i_0, i_1], \text{Range}[j_0, j_1]]]$	a submatrix with elements having row and column indices respectively in the ranges i_0 through i_1 and j_0 through j_1

Ways to get pieces of matrices.

Matrices in *Mathematica* are represented as lists of lists. You can use all the standard *Mathematica* list manipulation operations on matrices.

Here is a sample 3×3 matrix.

```
In[1]:= t = Array[a, {3,3}]
Out[1]= {{a[1, 1], a[1, 2], a[1, 3]},
          {a[2, 1], a[2, 2], a[2, 3]},
          {a[3, 1], a[3, 2], a[3, 3]}}
```

This picks out the second row of the matrix.

```
In[2]:= t[[2]]
Out[2]= {a[2, 1], a[2, 2], a[2, 3]}
```

Here is the second column of the matrix.

```
In[3]:= Map[\#[[2]]\&, t]
Out[3]= {a[1, 2], a[2, 2], a[3, 2]}
```

This picks out a submatrix.

```
In[4]:= t[[{1, 2}, {2, 3}]]
Out[4]= {{a[1, 2], a[1, 3]}, {a[2, 2], a[2, 3]}}
```

Here is another way to specify the submatrix, convenient when you deal with bigger matrices.

```
In[5]:= t[[Range[1,2], Range[2,3]]]
Out[5]= {{a[1, 2], a[1, 3]}, {a[2, 2], a[2, 3]}}
```

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